

~~Verification of~~ Row 2 and Row 3 have no leading entries and it's very easy to verify that the final matrix is in rref. $\therefore$ Row 2 and Row 3 have no pivot.

By Theorem 8 and $b \Leftrightarrow a$, we know that A is not an invertible matrix. $\square$

g) Suppose that A and B are $n \times n$ matrices and the equation $(AB)x = 0$ has a nontrivial solution. What can you say about the matrix AB ?

Solution: By the Invertible Matrix Theorem and $d \Leftrightarrow a$, we know that AB is not invertible.